

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim:- Performing linear regression analysis using lm() (R).

```
> wine_data <- read.csv("wine dataset.csv")
> model <- lm(Alcohol ~ Ash + Magnesium + Total.phenols + Flavanoids, data = wine_data)
> summary(model)

Call:
lm(formula = Alcohol ~ Ash + Magnesium + Total.phenols + Flavanoids,
    data = wine_data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.1478 -0.5496  0.1402  0.5446  1.7932

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.301801   0.588510   17.505  <2e-16 ***
Ash           0.381210   0.217659    1.751  0.0816 .
Magnesium     0.010492   0.004246    2.471  0.0144 *
Total.phenols 0.371019   0.182308    2.035  0.0434 *
Flavanoids    -0.049932   0.113701   -0.439  0.6611

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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7591 on 173 degrees of freedom
Multiple R-squared:  0.1454,    Adjusted R-squared:  0.1256
F-statistic: 7.357 on 4 and 173 DF,  p-value: 1.703e-05

> predicted_values <- predict(model)
> plot(wine_data$Alcohol, predicted_values,
+      main = "Actual vs Predicted Alcohol",
+      xlab = "Actual Alcohol",
+      ylab = "Predicted Alcohol")
> abline(0, 1, col = "red")
> wine_data <- read.csv("wine dataset.csv")
> wine_20 <- wine_data[1:20, ]
> model <- lm(Alcohol ~ Ash + Magnesium + Total.phenols + Flavanoids, data = wine_20)
> summary(model)

Call:
lm(formula = Alcohol ~ Ash + Magnesium + Total.phenols + Flavanoids,
    data = wine_20)

Residuals:
    Min       1Q   Median       3Q      Max
-1.00693 -0.17680 -0.02058  0.27558  0.58746

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.985296   1.660134   9.027 1.89e-07 ***
Ash          -0.989766   0.703844   -1.406  0.180
Magnesium     0.002639   0.012556    0.210  0.836
Total.phenols 0.204590   0.541539    0.378  0.711
Flavanoids    0.193361   0.452300    0.428  0.675

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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4651 on 15 degrees of freedom
Multiple R-squared:  0.233,    Adjusted R-squared:  0.02844
F-statistic: 1.139 on 4 and 15 DF,  p-value: 0.3758

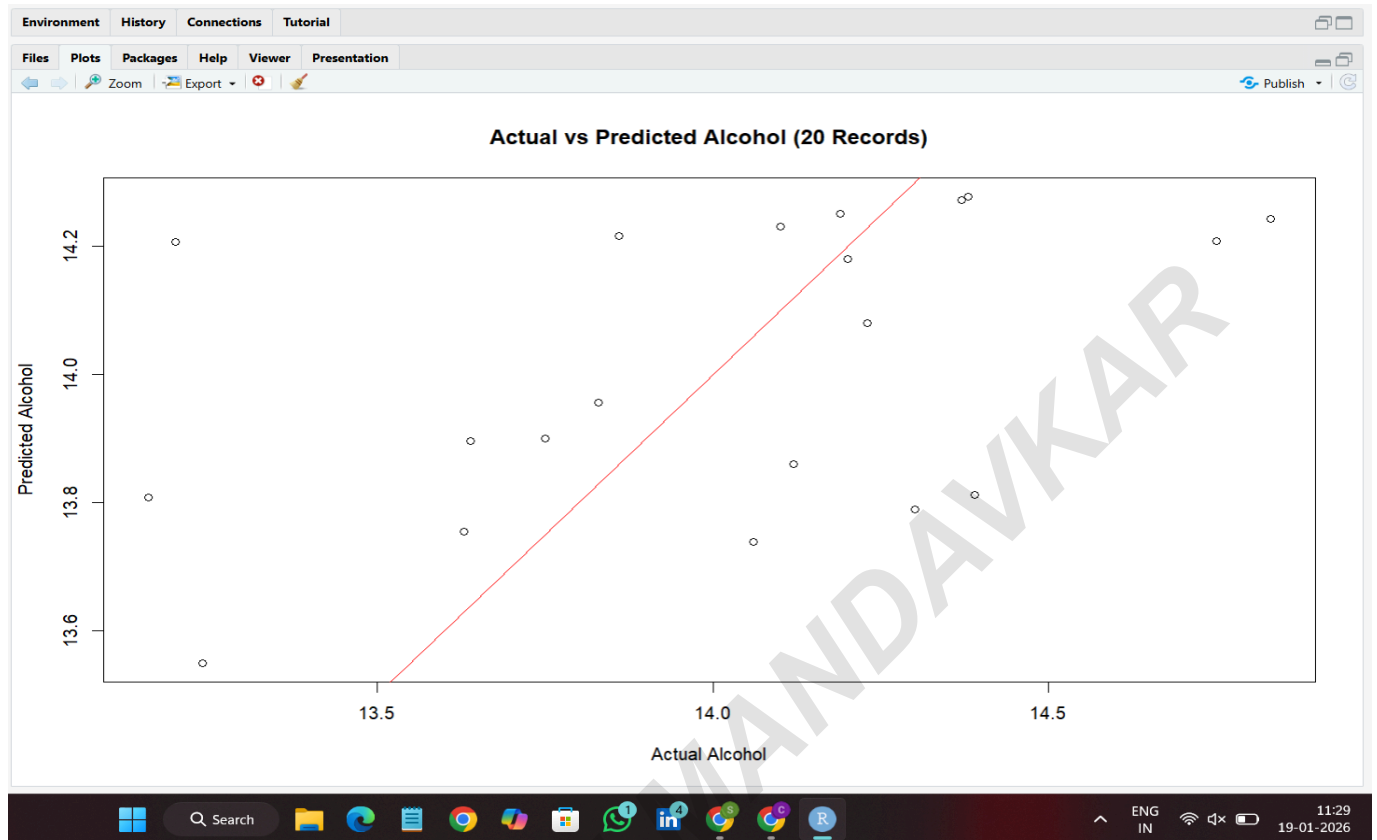
> predicted_values <- predict(model)
> plot(wine_20$Alcohol, predicted_values,
+      main = "Actual vs Predicted Alcohol (20 Records)",
+      xlab = "Actual Alcohol",
+      ylab = "Predicted Alcohol")
> abline(0, 1, col = "red")
>
```

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ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

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